

Indices Assessment

Q1.

Express as a single power of a

$$\frac{a^2}{\sqrt{a}}$$

where $a \neq 0$

Circle your answer.

a^1

$a^{\frac{3}{2}}$

$a^{\frac{5}{2}}$

a^4

(1)

Q2.

Identify the expression below that is equivalent to $e^{-\frac{2}{5}}$

Circle your answer.

$\frac{1}{\sqrt[5]{e^2}}$

$-\sqrt{e^5}$

$-\sqrt[5]{e^2}$

$\frac{1}{\sqrt{e^5}}$

(1)

Q3.

Simplify $\sqrt{a^{\frac{2}{3}} \times a^{\frac{2}{5}}}$

Circle your answer.

$a^{\frac{2}{15}}$

$a^{\frac{4}{15}}$

$a^{\frac{8}{15}}$

$a^{\frac{16}{15}}$

(1)

Q4.

(a) Write down the value of p and the value of q given that:

(i) $\sqrt{3} = 3^p$

(1)

(ii) $\frac{1}{9} = 3^q$

(1)

(b) Find the value of x for which $\sqrt{3} \times 3^x = \frac{1}{9}$

(2)

Q5.

- (a) Write $\sqrt[4]{x^3}$ in the form x^k .

(1)

- (b) Write $\frac{1-x^2}{\sqrt[4]{x^3}}$ in the form $x^p - x^q$.

(2)

Q6.

- (a) Write down the values of p , q and r given that:

(i) $8 = 2^p$;

(1)

(ii) $\frac{1}{8} = 2^q$;

(1)

(iii) $\sqrt{2} = 2^r$.

(1)

- (b) Find the value of x for which $\sqrt{2} \times 2^x = \frac{1}{8}$.

(2)

Q7.

- (a) Write down the values of p , q and r given that:

(i) $64 = 8^p$;

(ii) $\frac{1}{64} = 8^q$;

(iii) $\sqrt{8} = 8^r$.

(3)

- (b) Find the value of x for which

$$\frac{8^x}{\sqrt{8}} = \frac{1}{64}$$

(2)